



Intel labeled Hoff a "rock star" for his work on the microprocessor.

## Microchips

Early computers relied on transistors and other electrical parts connected **by hand**. This was a laborious process, and if any one of these connections broke, the whole bunch could fail. In 1958, American scientist Jack Kilby developed the integrated circuit (left). By 1961, these were a lot smaller and commonly known as **MICROCHIPS**. Each one consisted of hundreds of tiny parts, made from one piece of material (usually **silicon**). They made computer parts more reliable, organized, and consumed less power.

The Intel 8080 was used in the first commercial computers.

How it changed

Without microprocessors, we wouldn't have personal computers, or any of the smart appliances that help run our lives.

the world



The smallest wires in today's microprocessors are less than one-thousandth the width of a human hair.

## Minicomputer

In 1971, fellow American Ted Hoff was designing a new microchip for a scientific calculator for the company Intel. He thought it would be easier to make a chip that could be used for a **variety of functions**, as opposed to one that would work only for his calculator. His solution was the Intel 4004 microprocessor, **a minicomputer on a chip**. Further improvements led to the Intel 8080 chip, which came to be known as "the first truly usable microprocessor."

# Microprocessor

The tiny technology that is the **BIG BRAIN** in your computer

## Chips with everything

A microprocessor is like a **BRAIN**: It reads and adds to memory, carries out instructions, and communicates with other parts of the computer. Today's microprocessors power computers, phones, washing machines, and much more. They're thousands of times faster than the first ones, and yet they're small enough to **fit on a fingernail**.

